

Md. Raihan Siddik

Computer Vision | Vision-Language Models | Multimodal Learning | Efficient Deep Learning

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Profile

Electrical and Electronic Engineering graduate from Bangladesh University of Engineering and Technology (BUET), with a major in Communication and Signal Processing. My research interests include computer vision, vision-language models, multimodal learning, deepfake detection, industrial anomaly detection, autonomous systems, and efficient deep learning. I have experience developing models with PyTorch, conducting experiments, and testing model performance across different datasets. I am seeking PhD opportunities in computer vision and multimodal AI, especially in reliable vision-language systems for real-world applications.

Research Interests

Computer Vision; Vision-Language Models; Multimodal Foundation Models; Multimodal Learning; Efficient and Parameter-Efficient Deep Learning; Deepfake Detection and Media Forensics; Industrial Visual Anomaly Detection; Autonomous Visual Systems; Signal Processing; Trustworthy AI.

Education

Bangladesh University of Engineering and Technology (BUET), Dhaka

June 2026

B.Sc. in Electrical and Electronic Engineering –Major: Communication and Signal Processing

CGPA: 3.54 / 4.00 | Last two semesters: 3.84 / 4.00

Research Outputs

FaLoRA: Parameter-Efficient Frequency-Aware Adaptation for Generalizable Deepfake Detection

First-author manuscript

Manuscript in preparation for publication

CLIP-LoRA for Industrial Anomaly Detection

First-author research manuscript

Ongoing research; manuscript in preparation for publication

Research Experience

FaLoRA: Parameter-Efficient Frequency-Aware Adaptation for Generalizable Deepfake Detection

Undergraduate Thesis | Supervisor: Dr. S. M. Mahbubur Rahman, Professor, Department of EEE, BUET

- Proposed a frequency-aware LoRA framework combining DCT, FFT phase, and steganalytic noise filters to adapt a Vision Transformer with 0.94 million trainable parameters (approximately 2% of the model).
- Evaluated cross-dataset generalization without task-specific retraining, achieving AUCs of 90.96% on Celeb-DF v2, 79.34% on DFDC, 80.95% on WildDeepfake, and 89.26% on DeeperForensics.
- Designed the experimental protocol, implemented the PyTorch pipeline, conducted ablation studies, and analyzed frequency-domain and generalization behavior.

CLIP-LoRA for Industrial Anomaly Detection

Ongoing Research | Supervisor: Dr. S. M. Mahbubur Rahman, Professor, Department of EEE, BUET

- Investigating parameter-efficient adaptation of CLIP for industrial visual anomaly detection and defect recognition.
- Developing and evaluating a generalizable image-text framework that reduces trainable parameters while preserving representation quality and cross-category performance.
- Preparing the research for publication, including comparative experiments, ablation studies, and reproducible evaluation.

Selected Research and Engineering Projects

Real-Time Audio-Visual Zooming System

2026

IEEE Signal Processing Cup 2026

- Built a spatial audio separation system using DP-PCRN and Causal CRN with interaural phase and level difference features; achieved PESQ 3.05 and STOI 0.906.
- Optimized the model for mobile deployment using TensorFlow Lite int8 quantization.

UAV-Based Vegetation Stress Mapping for Tea Gardens

Robotics and Automation Project

- Developed an autonomous quadcopter-based visual analysis system using SegFormer for disease classification and vegetation-index heatmap generation, achieving 97.7% pixel accuracy.
- Integrated Python, OpenCV, Raspberry Pi, ArduPilot, and semantic segmentation for autonomous data collection and analysis.

Technical Skills

Programming: Python, C, C++, MATLAB

Deep Learning and Computer Vision: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, SegFormer, Vision Transformers, CLIP, LoRA

Data and Scientific Computing: NumPy, Pandas, signal processing, image and video preprocessing, evaluation metrics, experiment design

Tools and Platforms: Linux (Ubuntu), Git, GitHub, LaTeX, TensorFlow Lite, Raspberry Pi, ArduPilot

Research: Literature review, scientific writing, ablation studies, cross-dataset evaluation, reproducible research, technical presentation

Relevant Coursework

Artificial Intelligence and Machine Learning; Robotics and Automation; Digital Signal Processing; Random Signals and Processes; Probability and Statistics; Linear Algebra; Computer Programming; Introduction to Medical Imaging; Microprocessors and Embedded Systems.

Industrial Exposure

Bangladesh Submarine Cable Company Limited (BSCCL)

2025

International Internet Gateway Unit, Dhaka

Gained practical exposure to national-scale communications infrastructure, IP transit and peering, and network monitoring operations.

Languages

Bangla: Native | **English:** Fluent

References

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Dr. Md. Kawsar Alam

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